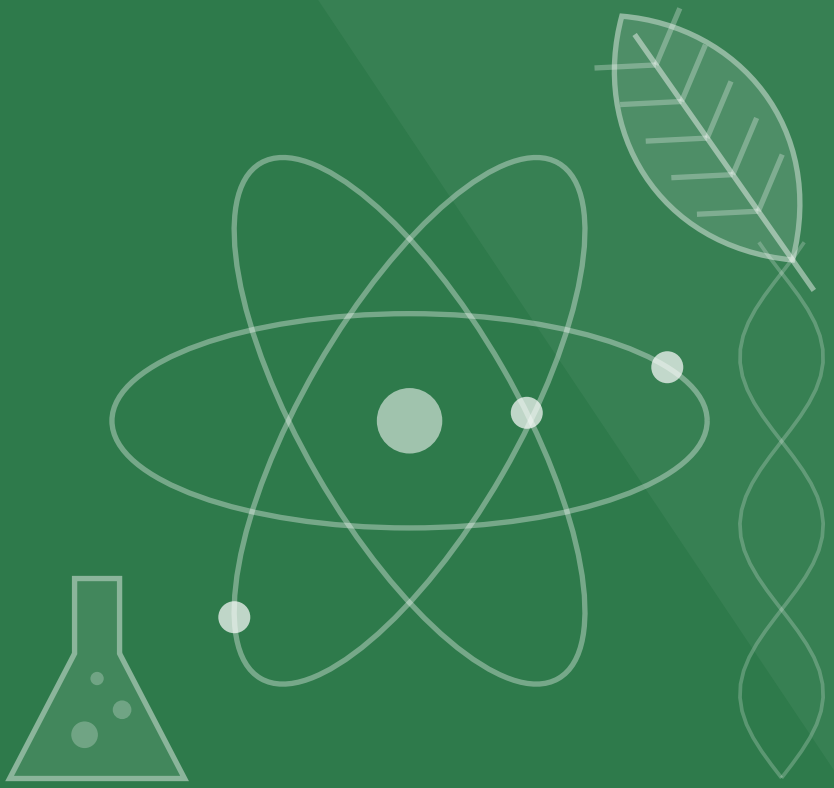
VENVLEARNING

STUDENT WORKBOOK



Science

GRADE 7

Daily practice, weekly checks, and unit tests

Unit 1: Cycles of Matter and Energy

VENVLEARNING

Grade 7
Science

STUDENT WORKBOOK

Unit 1 Cycles of Matter and Energy

A VenvLearning family homeschool curriculum

How to use this workbook

Every lesson in your textbook has matching practice pages here. Show your thinking in the work space. When a problem has an answer line, write your final answer there.

- Warm-Up reviews something you already know.
- Practice uses today's lesson.
- Think Deeper problems ask you to explain or find a mistake.
- Exit Ticket shows your teacher what you learned today.

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UNIT 1

Cycles of Matter and Energy

Science practices, Earth's spheres, the water, carbon, oxygen, and nitrogen cycles, photosynthesis, respiration, food webs, and biodiversity.

Your practice pages for Unit 1. Each lesson matches the same lesson number in your textbook.

UNIT VOCABULARY

independent variable	The variable that is changed.
dependent variable	The variable that is measured.
photosynthesis	Light + CO_2 + water → glucose + oxygen.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
nitrogen fixation	Changing N_2 into usable forms.
biodiversity	The variety of life in an area.

LESSON 1.1 · PRACTICE
Week 1 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.1: Variables, Data, and Scientific Explanations

Name _____ Date _____

TODAY'S GOAL

I can identify independent, dependent, and controlled variables and write a scientific explanation using CER.

Standards: SEP 3, 4, 6

WHAT TO DO

- [ ] Read Textbook Lesson 1.1 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Identify the Variables, Part D Analyze Data, Part E Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. Plant height is measured after different amounts of fertilizer. Independent variable?

2. Same experiment: dependent variable?

3. Name two controlled variables.

PART B Warm-Up

4 The variable you change

5 The variable you measure

PART C Identify the Variables

Directions: For the question, name the independent, dependent, and controlled variables.

6 Does the height of a ramp affect how far a car rolls? Independent:

7 Same: dependent:

8 Same: two controlled variables:

9 Write a hypothesis for the ramp experiment.

PART D Analyze Data

Directions: Use the data to answer the question. Describe the pattern and explain it.

10 Data: 0 g salt → water boils at 100.0°C; 20 g → 100.3°C; 40 g → 100.7°C. Write a CER.

PART E Exit Ticket

11 Why must controlled variables stay the same?

LESSON 1.2 · PRACTICE
Week 1 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.2: Earth's Spheres and the Water Cycle

Name _____ Date _____

TODAY'S GOAL

I can describe Earth's four spheres and model how energy from the sun and gravity drive the water cycle.

Standards: MS-ESS2-4

WHAT TO DO

- [ ] Read Textbook Lesson 1.2 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Practice, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. Which sphere includes clouds and air?

2. What process returns water vapor from plants?

3. What force makes rain fall and rivers flow?

PART B Warm-Up

Directions: Write the term that completes each line.

4 All living things make up the...

5 Water vapor cooling into droplets is...

PART C Practice

6 Which sphere does a river belong to?

7 Which sphere does a mountain belong to?

8 What provides the energy for evaporation?

9 What happens to water that soaks into soil?

10 Describe one interaction between two spheres in the water cycle.

11 Explain the droplets on the inside of the bag using the words evaporation and condensation.

PART D Exit Ticket

12 Name the two things that drive the water cycle and what each does.

WEEKLY CHECK
After Lesson 1.2

About 15 minutes

Weekly Check 1: Variables and the Water Cycle

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Name each variable, or answer the question.

1. Does the amount of sunlight affect bean plant growth? Independent variable:

2. Dependent variable:

3. One controlled variable:

4. Water vapor cooling into droplets is...

5. Which sphere includes all living things?

6. Name the two drivers of the water cycle and what each does.

LESSON 1.3 · PRACTICE
Week 2 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.3: The Carbon and Oxygen Cycles

Name _____ Date _____

TODAY'S GOAL

I can explain how carbon and oxygen cycle among living things, the atmosphere, the oceans, and rocks.

Standards: MS-LS2-3

WHAT TO DO

- [ ] Read Textbook Lesson 1.3 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Practice, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. Which process removes carbon dioxide from the air?

2. Name two processes that release carbon dioxide.

3. About what percent of air is oxygen?

PART B Warm-Up

Directions: Write the term for each description.

4 Coal, oil, and natural gas are...

5 Organisms that break down dead matter

PART C Practice

6 A tree takes in CO_2. Process?

7 A car burns gasoline. Process?

8 A fungus breaks down a log. Process?

9 You exhale. Process?

10 Explain why burning fossil fuels adds carbon to the atmosphere faster than natural processes remove it.

PART D Exit Ticket

11 How are the carbon and oxygen cycles connected?

LESSON 1.4 · PRACTICE
Week 2 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.4: The Nitrogen Cycle

Name _____ Date _____

TODAY'S GOAL

I can explain how nitrogen moves from the air into living things and back.

Standards: MS-LS2-3

WHAT TO DO

- [ ] Read Textbook Lesson 1.4 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Practice, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. What percent of air is nitrogen?

2. What does nitrogen fixation do?

3. Why do farmers plant legumes?

PART B Warm-Up

Directions: Write the term that completes each line.

4 Living things use nitrogen to build...

5 A bean plant is a...

PART C Practice

6 Can animals use nitrogen gas directly?

7 How do animals get nitrogen?

8 Name a non-living way nitrogen is fixed.

9 What returns nitrogen from dead organisms to soil?

10 Explain why rotating crops with legumes can help soil.

PART D Exit Ticket

11 Trace one nitrogen atom from the air into your body.

WEEKLY CHECK
After Lesson 1.4

About 15 minutes

Weekly Check 2: Carbon, Oxygen, and Nitrogen

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

1. Which process removes CO_2 from air?

2. Name a process that releases CO_2.

3. About what percent of air is nitrogen?

4. What do nitrogen-fixing bacteria do?

5. Why does burning fossil fuels change the carbon cycle?

LESSON 1.5 · PRACTICE
Week 3 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.5: Photosynthesis

Name _____ Date _____

TODAY'S GOAL

I can explain how plants use light energy to make sugar and oxygen from carbon dioxide and water.

Standards: MS-LS1-6

WHAT TO DO

- [ ] Read Textbook Lesson 1.5 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Analyze Data, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. Where does most of a tree's mass come from?

2. What are the products of photosynthesis?

3. What pigment captures light?

PART B Warm-Up

Directions: Write the answer for each item about photosynthesis.

4 Reactants of photosynthesis

5 Organelle where it happens

PART C Analyze Data

6 Disks floating after 10 min: lamp cup 9; dark cup 0. Which had more photosynthesis?

7 What gas made the disks float?

8 What was the independent variable?

9 Write a CER for the leaf disk results.

PART D Exit Ticket

10 Explain how photosynthesis transforms energy.

LESSON 1.6 · PRACTICE
Week 3 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.6: Cellular Respiration

Name _____ Date _____

TODAY'S GOAL

I can explain how cells release energy from food through cellular respiration.

Standards: MS-LS1-7

WHAT TO DO

- [ ] Read Textbook Lesson 1.6 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Compare, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. Where does cellular respiration happen?

2. What gas filled the balloon?

3. Do plants carry out respiration?

PART B Warm-Up

Directions: Write the answer for each item about cellular respiration.

4 Products of respiration

5 Reactants of respiration

PART C Compare

Directions: Write photosynthesis or cellular respiration for each question.

6 Which process stores energy in glucose?

7 Which process releases energy from glucose?

8 Which produces oxygen?

9 Which produces carbon dioxide?

10 Balloon A (sugar) grew to 28 cm; balloon B (no sugar) to 13 cm. Explain using CER.

PART D Exit Ticket

11 Explain how matter cycles but energy flows between photosynthesis and respiration.

WEEKLY CHECK
After Lesson 1.6

About 15 minutes

Weekly Check 3: Photosynthesis and Respiration

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

Show what you know

Directions: Write the answer for each item.

1. Products of photosynthesis

2. Products of cellular respiration

3. Organelle for photosynthesis

4. Organelle for respiration

5. Explain how photosynthesis and respiration cycle matter.

LESSON 1.7 · PRACTICE
Week 4 · Lesson 1 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.7: Food Webs and Energy Pyramids

Name _____ Date _____

TODAY'S GOAL

I can model how energy flows through a food web and explain why less energy reaches each higher level.

Standards: MS-LS2-3

WHAT TO DO

- [ ] Read Textbook Lesson 1.7 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Practice, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. Which way do food-chain arrows point?

2. If grass has 20,000 units, about how much reaches grasshoppers?

3. Why are top predators few?

PART B Warm-Up

Directions: Write the term for each description.

4 An organism that makes its own food

5 Organisms that break down dead matter

PART C Practice

6 Producers have 50,000 units. Primary consumers?

7 Secondary consumers?

8 Tertiary consumers?

9 In grass → rabbit → fox, the fox is a...

10 If a disease kills most rabbits, predict effects on the grass and the foxes. Explain.

PART D Exit Ticket

11 Explain the difference between how matter and energy move through an ecosystem.

LESSON 1.8 · PRACTICE
Week 4 · Lesson 2 of 2

Unit 1: Cycles of Matter and Energy

Lesson 1.8: Biodiversity and Ecosystem Services

Name _____ Date _____

TODAY'S GOAL

I can explain why biodiversity matters and evaluate ways to protect ecosystem services.

Standards: MS-LS2-5

WHAT TO DO

- [ ] Read Textbook Lesson 1.8 first.
- [ ] Do every part on these pages: Part A Check Your Understanding, Part B Warm-Up, Part C Practice, Part D Exit Ticket.
- [ ] Answer in complete sentences unless a line asks for one word.
- [ ] Check your work with your teacher and fix any mistakes.

PART A Check Your Understanding

Directions: Answer each question.

1. What is biodiversity?

2. Name two ecosystem services.

3. Why is a diverse ecosystem more stable?

PART B Warm-Up

Directions: Write the answer for each item.

4 An animal that carries pollen

5 Wetlands help prevent...

PART C Practice

6 Lawn square: 3 species. Field square: 14 species. Which has more biodiversity?

7 Explain one way the loss of pollinators would affect people.

8 Evaluate: A farmer wants to remove all hedgerows to plant more crops. Name one benefit and one cost to ecosystem services.

PART D Exit Ticket

9 Choose one ecosystem service and explain how people could protect it.

UNIT REVIEW
Study before the unit test

About 40 minutes

Unit 1 Review

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Practices and cycles

Directions: Write the term for each description.

1. The variable you measure

2. Plants releasing water vapor

3. Changing N_2 into usable forms

4. Burning fuel

PART B Energy

Directions: Answer each item. Show your math for the energy pyramid.

5. Photosynthesis equation (words)

6. Producers 30,000 units; primary consumers?

7. Why do energy pyramids narrow at the top?

UNIT TEST
At the end of the unit

About 45 minutes

Unit 1 Test

Name _____ Date _____

DIRECTIONS

- [ ] Work on your own. Don't use your textbook unless your teacher says you may.
- [ ] This test has 20 points.
- [ ] Read each passage and question carefully. Answer in complete sentences.
- [ ] When you finish, check that every question has an answer.

PART A Part A · Science practices (1 point each)

Directions: Name each variable for the experiment.

1. Does the temperature of water affect how much sugar dissolves? Independent variable:

2. Dependent variable:

3. Name one controlled variable.

PART B Part B · Cycles (1 point each)

4. Which process adds water vapor to the air from plants?

A condensation
B transpiration
C infiltration
D runoff

5. What force pulls precipitation down?

6 Which process removes carbon dioxide from the atmosphere?

A combustion
B decomposition
C photosynthesis
D respiration

7 Why can't animals use nitrogen gas directly?

8 Name the organisms that return nutrients from dead matter to soil.

PART C Part C · Energy in living things (1 point each)

9 Write the word equation for photosynthesis.

10 Write the word equation for cellular respiration.

11 Producers store 40,000 units of energy. About how much reaches secondary consumers?

12 Which is an ecosystem service?

A a parking lot
B wetlands filtering water
C a concrete channel

PART D Part D · Data analysis (4 points)

Directions: Use the data to answer each part. Give a claim, evidence, and reasoning.

13 Leaf disks floating after 12 minutes: bright light 10, dim light 5, dark 0. Write a CER explaining the results.

PART E Part E · Explain (4 points)

14 Use a food chain (grass → rabbit → fox) to explain how matter cycles and energy flows. Include photosynthesis, respiration, and decomposers.

Grade 7 Science

Student Workbook

A full year of Grade 7 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Workbook holds the daily practice pages for every lesson, plus a Weekly Check at the end of each week, the unit reviews, and the unit tests, with room to show thinking and write complete answers.

INSIDE THIS YEAR

01 Cycles of Matter and Energy

IN THIS BOOK

- Warm-ups, practice, and exit tickets for every lesson
- A Weekly Check for each week of lessons
- Unit reviews and unit tests
- Space to show work and explain thinking

THE COMPLETE SET



Textbook

Workbook

You are holding this book

Teacher Edition

Each lesson has the same number in all three books, so the textbook, workbook, and teacher pages always match.